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Comparison of Cohen's *d* and Hedges' *g* in One-Stage Meta-Analytic Structural Equation Modeling

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Comparison of Cohen's *d* and Hedges' *g* in One-Stage Meta-Analytic Structural Equation Modeling

Meta-analytic structural equation modeling (MASEM) has been used to illuminate psychological mechanisms by fitting structural equation models to meta-analyses. Standardized mean differences, used to represent meta-effects, are transformed into point-biserial correlations (r_{pb}) for MASEM. Though Hedges' *g* (*g*) is preferred over Cohen's *d* (*d*) to correct small-sample bias in individual studies, recent evidence suggests this may not extend to meta-analyses. Despite the widespread use of *d* and *g*, limited research has examined their impact on MASEM, particularly in meta-analyses of randomized controlled trials (RCTs). This study addresses this gap by comparing MASEM outputs – path coefficients and pooled correlations – based on r_{pb}s derived from *d* versus *g* using a formula assuming balanced control and treatment arms. With a dataset of RCTs on cognitive behavioral therapy's impact on youth depression through negative cognition, two one-stage MASEMs were conducted using r_{pb}s transformed from d or g. Given prior findings indicating that the difference in bias between *d* and g diminishes in meta-analyses with large samples, I hypothesized that the two MASEMs would yield similar path coefficients and pooled correlations when the average total sample size of studies exceeds 100. Indeed, results revealed negligible differences: both models produced similar pooled correlations and path coefficients. Findings indicate that choosing between *d* and *g* has minimally impacted MASEM outcomes, suggesting that researchers can compare and summarize effects across studies, regardless of whether they are *d* or *g*. Future research should investigate whether similar observations apply across various datasets, (distributions of) sample sizes, and base rate characteristics.

Introduction

The Role of Meta-Analyses and Standardized Mean Differences in Clinical Research

Meta-analyses have become increasingly prevalent among scientists (Fagard et al., 1996). One of the benefits of using meta-analyses in research is that they offer aggregating results from various (similar) studies to give a robust overview of the relationships between variables (Fagard et al., 1996). Moreover, these meta-analyses enable researchers to examine relationships akin to ones in the real world, where they facilitate the investigation of complex phenomena commonly studied in clinical (psychology) studies, primarily through randomized controlled trials (RCTs) (Weisz et al., 2023).

To integrate results across RCTs, a meta-effect is typically computed using standardized mean differences (SMDs) (Lin & Aloe, 2020). Standardization is done by dividing the difference between group means by the standard deviation. The result is a scale-free effect size that allows comparisons across studies that use different measurement units like the Beck Depression Inventory scores, seconds, milliseconds, etc. (Friedrich et al., 2008). As such, SMDs are scale-free representatives of the mean differences between control and treatment groups in RCTs (Hedges & Olkin, 1985). In RCTS, computing the mean difference is important since the randomization process requires a dichotomous variable (i.e., the assignment of control vs treatment), and the outcomes are (often) measured between baseline and follow-up time points on continuous scales (Vickers & Altman, 2001). Moreover, SMDs offer a standardized method to consistently interpret the differences between control and treatment groups when different studies use different measurements for similar variables (Hedges & Olkin, 1985). For instance, the effect of psychotherapy on depression symptoms can be measured with different depression scales, where standardizing the mean differences between the treatment and control groups offers a scale-free interpretation of the effect size of depression symptoms.

Effect Size Calculations

Nevertheless, there are various SMD estimators that researchers have the freedom to choose when measuring a meta-effect (Lin & Aloe, 2020). Cohen's *d* and Hedges' *g* are among the most popular effect sizes that can estimate SMDs (Marfo & Okyere, 2019). Even though there are various formulas to calculate Cohen's *d*, the formula that assumes that the comparison groups are independent of each other, in parallel with RCTs in this study, will be used:

$$d = \frac{\overline{X_1} - \overline{X_2}}{S_{\text{pooled}}}, \quad (1)$$

where $\overline{X}_1$ and $\overline{X}_2$ represent the sample means, and S_{pooled} is the pooled standard deviation between groups, and formulated as:

$$S_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}, \quad (2)$$

where n_1 and n_2 are sample sizes; S_1 and S_2 are standard deviations of Group 1 and Group 2, respectively (Marfo & Okyere, 2019). Since Cohen's *d* is shown to give biased estimations when the sample sizes are small, Hedges' *g* is commonly used to correct such biases and computed as:

$$g = d \times \left(1 - \frac{3}{4(n_1 + n_2) - 9}\right), \quad (3)$$

where *d* is Cohen's *d*, n_1 and n_2 are the sample sizes of Group 1 and Group 2, respectively (Hedges, 1981).

Since Cohen's *d* is shown to give biased estimations when the sample sizes are small, Hedges' *g* is commonly used to correct such biases. However, few studies have investigated how Cohen's *d* and Hedges' *g* differ in their impact on meta-analytic methods, particularly in meta-analytical structural equation modeling (MASEM) (Lin & Aloe, 2020).

MASEM

A frequent approach to analyzing complex patterns in meta-analysis is MASEM (Steinmetz & Block, 2022). This method allows researchers to investigate structural equation models using correlations among variables in multiple studies. Furthermore, researchers can extract the necessary correlations from various studies, even if those studies do not examine the exact same relationships (Steinmetz & Block, 2022). This approach is particularly suited for meta-analytic clinical psychology research, where mediation models – such as how Cognitive Behavioral Therapy (CBT) treats youth depression via negative cognition— are a common interest among researchers (Ng et al., 2023). For instance, researchers can compile correlations between CBT and negative cognition from various studies that might not specifically focus on youth depression while also collecting other associations between CBT and youth depression from different research that might not focus on negative cognition. Then, they can aggregate all these correlations from multiple studies for MASEM analyses.

As outputs, among many, MASEM provides pooled correlations and path coefficients within different stages (Jak & Cheung, 2024; Jak et al., 2021). A pooled correlation is the correlation between two variables averaged across multiple independent studies. A pooled correlation matrix, which MASEM produces in its first stage, is the matrix of averaged correlations between the variables of interest. In the second stage of MASEM, this pooled correlation matrix is used to test the theoretical relationship among variables using structural equation modeling (SEM), allowing researchers to estimate path coefficients and analyze how well the theorotical model fits the averaged data across studies. These path coefficients, estimated from the pooled correlation matrix, are standardized regression coefficients estimated in the SEM. They indicate how strongly one variable directly influences another after controlling for other variables in the model. Ultimately, first, MASEM requires individual correlations to aggregate them into one matrix and then to estimate path coefficients.

Converting SMDs into Point-Biserial Correlation Coefficients

Although MASEM requires correlation coefficients as input, most studies increasingly report SMDs, such as Cohen's *d* and Hedges' *g*, instead of correlation coefficients (McGrath & Meyer, 2006). Thus, Cohen's *d* and Hedges' *g* must be transformed into (point-biserial) correlation coefficients. While Pearson correlation coefficients are used for continuous variable pairs, the point-biserial correlation is used to describe relationships between dichotomous variables (e.g., treatment vs control) and continuous variables (e.g., depression score) (De Jonge et al., 2020). This study focuses on transforming SMDs into point-biserial correlations since it is the appropriate correlation type for dichotomous variables, and the mediation model of this study has a dichotomous variable (see Figure 1). As previously noted, Cohen's *d* and Hedges' *g* may yield different results. While several studies have examined the differences between Cohen's *d* and Hedges' *g* within individual studies, limited research has explored how these differences impact their conversion to correlation coefficients and how these coefficients influence the MASEM outputs, such as pooled correlations or path coefficients. The choice of SMD, subsequently, may affect the model estimates in MASEM, particularly in the context of RCTs. Hence, this study aims to bridge this gap in the literature by looking at the effects of Cohen's *d* and Hedges' *g* when they are transformed into correlation coefficients and inputted into MASEM for meta-analyses of RCTs.

There are multiple conversion formulas to transform SMDs into correlation coefficients, and each formula works optimally under specific conditions concerning group sizes and balance between two groups (Aaron et al., 1998; Borenstein et al., 2009; Hedges & Olkin, 1985; McGrath & Meyer, 2006). Hedges & Olkin's (1985) formula is recommended for general purposes:

$$r_{pb} = \frac{d}{\sqrt{d^2 + \frac{N^2 - 2(n_1^2 + n_2^2)}{n_1 n_2}}}, \tag{4}$$

where d is the SMD (of choice), N is the total sample size, and n_1 and n_2 are the sample sizes of Group 1 and Group 2, respectively. When N approaches infinity (i.e., when large sample size approximations hold), this formula diverges into Borenstein et al.'s (2009) large-sample formula:

$$r = \frac{d}{\sqrt{d^2 + a}}, \tag{5}$$

where the term a represents a correction when $n_1 \neq n_2$:

$$a = \frac{(n_1 + n_2)^2}{n_1 n_2}. \tag{6}$$

On the other hand, when groups are balanced (i.e., $n_1 = n_2$), Aaron et al. (1998) recommend their equal-samples formula to transform SMDs into point-biserial correlations:

$$r = \frac{d}{\sqrt{d^2 + 4 - \left(\frac{8}{N}\right)}}, \tag{7}$$

where r denotes the (point-biserial) correlation, d represents the SMD (i.e., Cohen's d or Hedges' g in this study), and N is the total sample size.

Among the above-mentioned transformation formulas, I decided not to use the large-sample formula (Equation (5)) because it is intended for large sample sizes (i.e., larger than a "few hundred"), whereas the average total sample size of this study's dataset is approximately 101 (Cooper et al., 2019, pp. 45-46). Instead, I chose the equal-samples formula (Equation (7)) due to a fundamental difference between SMDs and point-biserial correlations. While Cohen's d and Hedges' g remain stable regardless of group (im)balance, point-biserial correlation is sensitive to base rates (i.e., the proportion of participants in each group) (McGrath & Meyer, 2006). When the control and treatment groups are unbalanced, point-biserial correlation

systematically decreases, resulting in inconsistent effect size comparisons across studies (McGrath & Meyer, 2006; Ng et al., 2023). For instance, if an outcome is rare in the population (e.g., diagnosis of conduct disorder), randomly splitting participants into groups might produce samples with equal group sizes (i.e., a base rate of 0.50). However, this does not reflect the true population prevalence, which is unlikely to be 0.50. Such mismatches can distort the point-biserial correlation, as it is sensitive to group proportions. Then, this study considers groups to be balanced because RCTs typically use equal (random) allocation to make groups as balanced as possible and prevent systematic differences between groups and several biases associated with it (Altman & Bland, 1999). Since this study focuses on RCTs, assuming balanced groups helps eliminate the influence of unequal group sizes on the correlation estimates in meta-analytical frameworks such as MASEM.

Aim of the Study and Hypothesis

Given the gap in the literature, this study aims to investigate how the equal-samples conversion formula (Equation (7)) for transforming SMDs into point-biserial correlation coefficients differs in terms of pooled correlations and path coefficient estimates when either Cohen's *d* or Hedges' *g* is used as input. This comparison is conducted within the context of a mediation model (examining the effect of Cognitive Behavioral Therapy on youth depression via negative cognition) using two one-stage MASEMs. Based on the results, researchers can be cautious when they include different effects in their meta-analyses, as they need to determine whether to compare and summarize these different effects as similar, especially when they are derived from Cohen's *d* and Hedges' *g*.

Although Hedges' *g* is usually preferred over Cohen's *d* to correct small-sample bias in individual studies, Lin and Aloe (2020) showed through simulation studies that this correction may not be observable in meta-analyses. In fact, Hedges' *g* may exhibit greater bias than Cohen's *d*, especially when sample sizes are small to moderate – approximately 30-70 (Lin & Aloe, 2020). Furthermore, Lin (2018) reached a similar conclusion and demonstrated that Cohen's *d* gives less biased results than Hedges' *g* in meta-analysis with small sample sizes (i.e., 5-10 participants per study). However, this difference in bias between the two SMDs diminishes as the sample size increases, becoming minor after about 50 participants per study and practically negligible as the sample size approaches 100 (see Figure 2 in Lin, 2018).

Still, both Lin and Aloe's (2020) and Lin's (2018) studies only focused on what happens to SMDs when they are pooled to compute a meta-effect in meta-analyses. On the other hand, within my study's context, these SMDs are first transformed into point-biserial correlation coefficients, and then those correlation coefficients are pooled to be fed into a MASEM. Despite these differences, a meta-effect is computed in either scenario: either as pooled SMDs in meta-analyses or pooled (point-biserial) correlation coefficients in MASEM. Hence, it is reasonable to examine whether the differences observed between Cohen's *d* and Hedges' *g* in meta-analytic contexts also emerge in the context of MASEM analyses. Then, I hypothesize that pooled correlations and path coefficients estimated from one-stage MASEM will not differ when calculated using Cohen's *d* and Hedges' *g* when the average total sample size of the studies exceeds 100.

Interpreting Differences Between Models

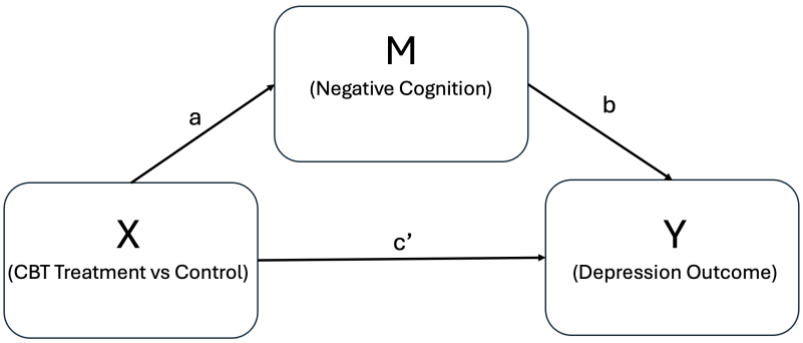
After conducting two MASEM models based on point biserial correlations based on Cohen's *d* and Hedges' *g*, I will apply Funder and Ozer's (2019) guidelines for interpreting effect sizes to assess whether the two models are meaningfully different. According to these guidelines, an effect size of .05 shows a "very small" effect for individual observations (Funder & Ozer, 2019, p. 166). Still, it can become meaningful over time (e.g., long-term effects of CBT on depression outcome) or when effects accumulate over many instances. Likewise, an effect size of .10 signifies a small effect for single observations but can quickly become meaningful, especially in meta-analyses that integrate data from diverse research. An effect size around 0.20 is considered practically meaningful even in short-term effects (e.g., immediate relief in depressive symptoms following an intervention), while a large effect size (r = .30) is meaningful in both short-term and long-term contexts. Lastly, an effect size around .40 reflects a substantially large effect size, which has the potential of being an overestimation in large samples or hard-to-replicate. In the end, if the pooled correlations of the two models fall within the same interpretative range, and path coefficients are both numerically and directionally similar, I will consider the two models as similar.

Selected Mediation Model and Outline of the Study

The dataset used in this study consists of meta-analytical data that was collected from three decades of RCTs to uncover how CBT and Interpersonal Psychotherapy (IPT) influence

youth depression through mediation effects (Ng et al., 2023). This study will focus on a subset of the dataset, focusing on how CBT affects youth depression through negative cognition, with variables being CBT (dichotomous: treatment vs control), negative cognition (continuous), and depression outcome (continuous). I will investigate a partial mediation model, as increased engagement in pleasant activities was also a significant mediator in some studies (Ng et al., 2023). Moreover, when Ng et al. (2023) also applied MASEM to the same mediation model, they found that the indirect effect of the treatment condition was significant. Hence, a similar significant indirect effect is also expected in this study. See Figure 1 for the visualization of the mediation model.

Figure 1. *Mediation Model of CBT Treatment, Negative Cognition, and Depression Outcome*



Note. Path *a* represents the effect of CBT treatment (*X*) on negative cognition (*M*). Path *b* represents the effect of negative cognition (*M*) on depression outcome (*Y*). Path *c'* is the direct effect of CBT treatment (*X*) on depression outcome after controlling for negative cognition (*M*).

In the remainder of this paper, the methods section will provide details about the dataset, the transformation process used to answer the research question, and the execution of 2 one-stage MASEM models to compare SMDs. The results section will compare the models based on path coefficients. Then, the discussion section will interpret the findings, discuss the implications of SMD selections, and examine the strengths and limitations of the study. Finally, the conclusion will outline future research and summarize key points.

Methods

Dataset and Variables

The dataset was drawn from Ng et al.'s (2023) meta-analysis, which aimed to answer how cognitive behavioral therapy and interpersonal psychotherapy improve depression in

youngsters. They utilized MASEM to investigate seven candidate mediators between the type of therapy and youth depression. Although their dataset included seven candidate mediators, the selected part of the dataset for this study focused on the effects of CBT on youth depression, using negative cognition as the mediator (Ng et al., 2023). Since the meta-analytic dataset synthesizes the results of multiple studies, no participant data was present in the original dataset, so no privacy violation was observed.

The variables of interest in this study were CBT treatment (dichotomous: treatment vs control), negative cognition (continuous), and youth depression outcome (continuous). Here, in line with Ng et al.'s (2023) reporting, a high score in negative cognition and depression outcome refers to more favorable outcomes such as less frequency of negative cognition or less depression severity. Furthermore, the dataset included parameters to calculate SMDs (i.e., mean, standard deviation, and sample size for treatment and control groups) for the relationships between binary and continuous variables. However, the dataset did not report these parameters to calculate SMD when the relationship was between continuous variables (i.e., between negative cognition and depression outcome). Instead, it reported Pearson correlation coefficients in such relationships. Hence, for the relationships that include dichotomous variables (CBT), SMDs in the form of Cohen's *d* and Hedges' *g* were calculated with Equations (1) and (3), respectively. Then, these values were converted into point-biserial correlation with equal-groups conversion (Equation (7)) before being inputted into one-stage MASEMs. For the relationship between negative cognition and depression outcome, the reported Pearson correlation coefficients were used as input to one-stage MASEMs.

In the end, two one-stage MASEM were conducted: one using the point-biserial correlation coefficients transformed from Cohen's *d* and the other from Hedges' *g*. Later, the results of these models (particularly pooled correlations and path coefficients) were compared to reveal whether Hedges' *g* and Cohen's *d* differ in the MASEM context. The comparison of pooled correlations was done based on Funder and Ozer's (2019) study, which gives guidelines for interpreting effect sizes in psychological research. The path coefficients were compared based on their directionality and magnitude.

Aggregation of Effect Sizes and Sample Sizes

In the meta-analytic dataset, dependent effect sizes – multiple effect size estimates for the same relationship within the same study – exist. These studies measured youth depression

outcomes and negative cognition with different measurements or instruments. For instance, in a study within the dataset, youth depression was measured by different inventories (e.g., Beck Depression Inventory and Bellevue Index of Depression) and persons of interest (e.g., child, parent, and teacher). If not handled, these dependencies result in systematic biases. For instance, for depression in youth, selecting only one aspect of measurement (such as only children's self-report questionnaires) could yield a systematic bias where other people of interest (such as parents) are ignored (Cohen et al., 2019).

Bilici et al. (2025) pointed out six strategies to handle these dependencies: simple averaging, weighted averaging, random elimination, largest effect size selection, ignoring dependency, and the Wilson-Polanin-Lipsey approach. Among them, aggregation methods (simple and weighted averaging) worked best to decrease the bias in standard errors. Thus, in this study, the aggregation was done by computing the simple mean of standardized effect sizes whenever there were multiple measurements per one relationship. Then, the (aggregated) effect sizes were transformed into point-biserial correlation coefficients to be inputted into MASEMs.

Furthermore, a similar issue was present for the sample sizes. Due to (unexpected) dropouts or other reasons that might affect participant counts, several measurements belonging to the same study had different sample sizes. Since MASEM requires the sample sizes of the studies within its input, likewise with the effect sizes, these sample sizes were averaged via simple averaging.

Data Preprocessing

Prior to conducting two one-stage MASEMs (via correlation coefficients derived from Cohen's *d* and Hedges' *g*), processing was done to create two MASEM-compatible datasets. The path coefficient estimates of the results were compared to investigate the differences between Cohen's and Hedges' g. All the preprocessing was handled using R (version 4.4.2; R Core Team, 2024). The R code for computing Cohen's *d* and Hedges'*g*, as well as the Lavaan syntax, can be found in Appendices A, B, and C, respectively.

The dataset used in this study belongs to Ng et al.'s (2023) study, which examined several potential mediators of the relationship between CBT and depression in youth, including negative cognition. All files, including the datasets used in this study, are publicly available on the Open Science Framework's (OSF) webpage, accessible via https://osf.io/xpku2/ (Ng, 2023).

To avoid any potential confusion, all column names and their corresponding values mentioned in this section are presented in quotation marks, exactly as they appear in the original dataset.

Note that, initially, two scripts were created. One calculated and reported results based on Cohen's *d* and the other based on Hedges' *g* (see Appendices A and B, respectively). All the other operations were precisely the same for both of the scripts.

First, two datasets, ESData.csv and unimeta.csv, were imported, which included raw data of all the pair-wise relationships among variables. Second, IDs of primary studies related to the CBT study (i.e., the column "IPT" having the value "0") were extracted from the data frame that included univariate study-level analysis (i.e., unimeta.csv) (Ng et al., 2023). This resulted in 28 primary studies (among 34) that focused on the mechanisms of CBT.

Third, all the relevant parameters of primary studies were extracted from the Esdata.csv, which included all the data required to compute Cohen's *d* (via Equation (1)) and Hedges' *g* (via Equation (3)). Here, in Esdata.csv, columns "var1type" and "var2type" represent the codenames for variables, "MA" being negative cognition, "O" being the outcome variable (i.e., youth depression), and "1" is the treatment variable (i.e., CBT treatment). The column "r" provides the Person's product-moment correlation. When the relationship in Esdata.csv is between a binary and a continuous variable, the value of "r" represents Ng et al.'s (2023) already-transformed correlation coefficient from the SMDs. If it is between two continuous variables (i.e., negative cognition and youth depression), it represents Pearson's product-moment correlation. Only in the latter case, this correlation was used in the final analysis. All the already-transformed correlation coefficients were ignored since I used a transformation formula different from Ng et al. (2023). For the rest of the relationships (i.e., the dichotomous relationships: treatment and mediator, treatment and outcome), Cohen's *d* and Hedges' *g* were computed using the values from the columns: "mean_t," "mean_c," "n_t," "n_c," "SD_t," "SD_c," representing the means, sample sizes, and standard deviations of treatment and control group, respectively.

Filtering was applied to capture the post-treatment scores (but not the pre-treatment, for instance). In the dataset, the columns "var1time" and "var2time" represent the time points of the measurements. Only the post-treatment scores of the mediator (i.e., negative cognition) and the outcome (i.e., depression outcome) were selected. The treatment condition (i.e., CBT) had no time point as the variable indicated whether a person was in the experimental or control group). Table 1 shows how mediation paths were identified based on the variable type and measurement

timing. Each row represents a path in the mediation model, defined by the type and measurement time of the involved variables. For instance, path *a* captures the effect of CBT (treatment variable) on negative cognition (mediator), measured from baseline selection of treatment versus control (i.e., Var1time = "0") to pos-treatment (i.e., Var2time = "3"). At the end of the filtering, negative cognition and depression outcomes were measured at posttreatment, and CBT had no time point as the variable indicated whether a person was in the experimental or control group). The timing was selected this way to ensure capturing the (possible) effect of CBT treatment on depression outcome (via negative cognition) in parallel with the research question. This filtering based on the variable types and measurement times resulted in 278 bivariate relationships in 28 studies.

Table 1. *Extracted mediation paths based on variable type and measurement time*

Path	Var1type	Var2type	Var1time	Var2time
a	1	MA	0	3
b	MA	O	3	3
c'	1	O	0	3

Note. For "Var1type" and "Var2type", "1" indicates the treatment variable (CBT), "MA" the mediator (negative cognition), and "O" the outcome (depression). Regarding "Var1time" and "Var2time", value "0" represents the baseline time point (e.g., treatment vs control selection), and "3" represents the post-treatment time point. Path a represents the CBT-negative cognition variable pair; b represents the negative cognition-depression outcome pair; and c' represents the CBT-depression outcome pair. Refer to the source code for the implementation in R (See Appendices A and B).

Fourth, Cohen's *d* and Hedges' *g* were calculated via Equation (1) and (3), respectively. This calculation is the only difference between the two scripts (see Appendices A and B). After the SMDs were calculated, their sign was reversed based on the values of the column "NumOppDir." It represents the number of variables in the opposite direction in a bivariate relationship. Here, each score has an expected direction. When a higher score in the measurement means higher severity of symptoms, this measurement is in the expected direction. If not, the measurement is labeled as "opposite direction." However, "NumOppDir" only represents the number of opposite measurements in a relationship between two variables. Hence, the computed SMD's sign was reversed only if "NumOppDir" had the value of "1" (i.e., only one of the measurements was in the opposite direction). No reversal was needed if it had the values of "0" or "2" since both measurements were in the same direction. This reversal of the

sign rule was also applied to the “r” column that includes either the already-transformed or reported values of correlation coefficients (Ng et al., 2023).

Fifth, another sign reversal was applied to SMDs and correlation coefficients. It is based on three arbitrary rules of Ng et al.’s (2023) in their preprocessing code for effect sizes in EScomputation.R, accessible via https://osf.io/xpku2/files/ (Ng, 2023). While manually analyzing the directional meanings of batteries reported in each study, they identified that some signs needed to be reversed again, which they captured with three rules in their code. For my study, only one rule was important: If the relationship in the data was between CBT (i.e., “varType” having the value of “1”) and outcome (i.e., “varType” having the value of “0”), signs of SMDs and correlation coefficients were reversed again. In the end, higher scores on the depression outcome and negative cognition variables represent more desirable outcomes, indicating lower depression severity and less frequent (and/or intensity of) negative cognition (Ng et al., 2023).

Sixth, in parallel with the previously mentioned Bilici et al.’s (2025) results, different (and dependent) SMDs, correlation coefficients, and sample sizes that belonged to the same study ID were aggregated via simple averaging. During the averaging computations, missing values were omitted. Following the aggregation, each study included one sample size, two SMDs corresponding to the two binary-continuous (CBT and negative cognition, CBT and depression outcomes), and one correlation coefficient reflecting the continuous-continuous relationship (negative cognition and depression outcomes). The aggregation resulted in a total of 66 independent bivariate relationships focused on 28 studies.

Then, both Cohen’s *d* and Hedges’ *g* were converted to point-biserial correlations via equal-groups conversion (Equation (7)). After, all the rows that belong to the same study ID were collapsed into one row, making the final data frame with 28 rows (corresponding to 28 studies) and WebMASEM-compatible (Jak, 2023). Since not all studies investigated all three variables, the missing values were left as “NA” values in R, as WebMASEM (i.e., the web tool that was used to conduct MASEMs) can handle them (Jak et al., 2021). In the final dataset, 23 relationships belonged to CBT treatment and negative cognition, which can be found in the column labeled "M_X." 28 belonged to the outcome of depression resulting from CBT treatment, specified in column "Y_X." Furthermore, there were 15 relationships that linked negative cognition to the depression outcome, shown in column "Y_M." Ultimately, two scripts were run

to create two files. One file contains the point-biserial correlations derived from Cohen's d, while another contains the point-biserial correlations derived from Hedges' g.

MASEM Analyses & Findings

After creating two datasets, these data frames were formatted according to the given input structure by the WebMASEM Shiny app, reachable through https://osf.io/wh6d3/ (Jak, 2023; Jak et al., 2021). WebMASEM app presents a web interface where users can conduct one-stage MASEMs and two-stage SEMs (TSSEMs) without using explicit coding in R. After creating the WebMASEM-compatible Excel files in the comma-separated values (CSV) format, the mediation model was specified via a lavaan syntax that captures the mediation model with an expected indirect effect (see Appendix C; Rosseel, 2012). Since the created output files are in CSV format, the separator was selected as "Comma" in the WebMASEM website. There were 3 variables in total, and their names were given as "X," "M," and "Y," the columns that include correlations were selected as "M_X," "Y_X," and "Y_M," and the column that includes sample size was chosen as "N." No moderator was involved in the analysis, so no selection was made. Lastly, default arguments of WebMASEM were used for the analysis except for the starting value to calculate random-effect variances. This parameter was changed from the default value of 0.05 to 0.02, as the default value resulted in unrealistic standard errors within 45 and 116. When the parameter was set to 0.02, the standard errors were adjusted and inherited values between 0.01 and 0.04.

Then, the datasets were analyzed using WebMASEM to perform one-stage MASEM. Although WebMASEM conducts a one-stage MASEM, it can also conduct TSSEM and allows the user to explore intermediary outputs of the one-stage MASEM. First, two pooled correlation matrices (one per model) were extracted from this first stage. Second, from the outputs of one-stage MASEM, six path coefficients (three per model) were extracted. In the end, there were two models. In the first model, point-biserial correlations were derived from Cohen's *d* using Aaron et al.'s (1998) equal-groups conversion formula (Equation (7)). In the second model, this formula was applied to Hedges' *g*. The goal was to evaluate whether the choice of effect size estimator (i.e., Cohen's *d* vs. Hedges' *g*) influences the resulting path coefficients and/or pooled correlations after the conversion to point-biserial correlations in a meta-analytical context. Finally, Funder and Ozer's (2019) guidelines for effect size interpretations were used to assess whether the results of the two models' pooled correlations were meaningfully different. With

respect to path coefficients, interpretation was done based on the directionality and the magnitude of the path coefficients.

Results

Research Question and Data Analysis

This study explored whether using Cohen's *d* versus Hedges' *g* (as input for point-biserial correlations) differs in the outcomes of MASEM when the sample sizes exceed 100. To address the research question, effect sizes were derived from Ng et al.'s (2023) meta-analytical dataset, which investigated the mediation effect of negative cognition between CBT and youth depression outcome among 28 studies, with an average sample size of approximately 101. SMDs of dichotomous-continuous relationships were converted to point-biserial correlations using the equal-groups formula (Equation (7)), and continuous-continuous relationships used reported Pearson's correlation coefficients. Two MASEMs were conducted: one using Cohen's *d*-based point-biserial correlation coefficients and the other using Hedges' *g*-based point-biserial correlation coefficients. The overall results of the two models – particularly pooled correlation coefficients and path coefficients – can be found in Table 2.

Table 2. *Path coefficients and pooled correlations of Cohen's d- and Hedges' g-based MASEMs*

Path	Pooled Correlations		Path Coefficients	
	Cohen's d-based MASEM	Hedges' g-based MASEM	Cohen's d-based MASEM	Hedges' g-based MASEM
a	0.13*	0.13*	0.13*	0.13*
b	0.40*	0.40*	0.39*	0.39*
c'	0.12*	0.12*	0.07*	0.07*

Note. Path *a* represents the CBT-negative cognition variable pair, b represents the negative cognition-depression pair, and *c*' represents the CBT-depression pair. * $p < .001$.

Findings: Pooled Correlations

In the model using point-biserial correlation coefficients based on Cohen's d, all the relationships between variables (as depicted in Figure 1) were significantly correlated. The

pooled correlation between CBT and negative cognition was positively significant (r_{MX} = 0.13, 95% CI [0.09, 0.17], p < .001), indicating that receiving CBT treatment is positively correlated with more favorable negative cognition outcomes (i.e., experiencing fewer or less intensity of them). The pooled correlation between negative cognition and depression outcome was also significant (r_{MY} = 0.40, 95% CI [0.31, 0.49], p < .001), suggesting that a lower frequency (and/or intensity) of negative cognitions is positively associated with better depression outcomes (i.e., less depression outcome). Finally, CBT treatment was correlated significantly and positively with depression outcome (r_{YX} = 0.12, 95% CI [0.08, 0.17], p < .001), showing that undergoing CBT treatment is related to less severity of depression.

For the model using Hedges' *g*-based point-biserial correlation coefficients, all variable pairs were significantly and positively associated. The significant pooled correlation between CBT and negative cognition (r_{MX} = 0.13, 95% CI [0.09, 0.17], p < .001) suggests that CBT was linked to more favorable negative cognitions. The pooled correlation between negative cognition and depression outcome was significant as well (r_{MY} = 0.40, 95% CI [0.31, 0.49], p < .001), implying that lower levels of negative cognition tend to coincide with better depression outcomes (i.e., lower severity). Finally, the significant pooled correlation between CBT treatment and depression outcome (r_{YX} = 0.12, 95% CI [0.07, 0.17], p < .001) demonstrates that patients who received CBT also experienced less severe depression severity.

Findings: Path Coefficients

First, the one-stage MASEM based on Cohen's *d* estimated a significant positive effect of CBT treatment on negative cognition (β_{MX} = 0.13, p < .001). In other words, compared to control conditions, CBT resulted in better negative cognition outcomes (i.e., less frequent and/or less intensive outcomes). Moreover, the direct effect of CBT treatment on depression outcome was also a positive significant effect (β_{YX} = 0.07, p < .001), meaning that CBT relieved depressive symptoms. Then, MASEM estimated that improvements in negative cognition significantly improved depression outcomes (β_{YM} = 0.39, p = < .001). Furthermore, the indirect effect of CBT on depression outcome via negative cognition was significant (β_{MX} x β_{YM} = 0.05, 95% CI [0.03, 0.07]), inferring that the mediating effect of negative cognition was present. Note that although WebMASEM did not report the p-values of the indirect effect, since the 95% confidence interval

did not include the value 0, the indirect effect was significant. Since all effects were significant, a partial mediation was also observed in the mediation model (see Figure 1).

Second, when Hedges' *g* was used as the effect size, one-stage MASEM computed a significant positive effect of CBT on negative cognition ($\beta_{MX} = 0.13$, $p < .001$). Similarly to Cohen's d, the direct effect of CBT treatment on depression outcome was positive and significant ($\beta_{YX} = 0.07$, $p < .001$). Also, negative cognition affected depression outcomes positively and significantly ($\beta_{YM} = 0.39$, $p < .001$). Moreover, the indirect of CBT treatment on depression outcome via negative cognition was positive and significant ($\beta_{MX} \times \beta_{YM} = 0.05$, 95% CI [0.03, 0.07]). Like the previous model, partial mediation was observed as all the effects were significant. Then, in this model, compared to the control condition, people who received CBT reported fewer depressive symptoms directly and/or via experiencing fewer (or less extensive) negative cognitions.

Findings: Contrasting the Models

In parallel with Funder and Ozer's (2019) guidelines for interpreting effect sizes, the observed effect sizes in both models ranged from small to very large. The pooled correlations between CBT treatment and depression outcome, as well as CBT treatment and negative cognition, remained small-to-moderate, with correlations of approximately 0.13. This suggested that these correlations represent meaningful effects in practice, especially considering the long-term impact of psychological interventions (such as CBT) and the diversity of the meta-analysis, which this study used as a primary data source (Funder & Ozer, 2019). Whereas the pooled correlations between negative cognition and depression in both models were "very large" as they were approximately 0.40, suggesting a practically meaningful but potentially overestimated and hard-to-replicate relationship within the context of psychological research (Funder & Ozer, 2019, p. 156).

Comparing the two models, one with Cohen's *d* and the other with Hedges' *g*, the pooled correlations and path coefficients differed only marginally, with absolute differences ranging from 0.0003 to 0.0013 for pooled correlations and 0.0005 to 0.0014 for path coefficients. According to the effect size interpretation guidelines of Funder & Ozer (2019), pooled correlations of both models fell within the same interpretative range (i.e., either they were both small or they were overly large). Regarding the path coefficients, minor differences at the third

or fourth decimal place do not alter their practical interpretation, especially when they all were positively and statistically significant. Moreover, identical patterns of significance and directionality were observed across both models, supporting the conclusion that decimal differences in either pooled correlations or path coefficients are practically negligible. These findings suggested that Cohen's *d* and Hedges' *g,* when transformed to point-biserial correlation coefficients with equal-groups conversion (Equation (7)), give practically equivalent results in one-stage MASEM in the context of a mediation model with three variables.

Discussion

This study examined whether the choice between Cohen's *d* and Hedges' *g* yields different outcomes when they are transformed into point-biserial correlations and fed into one-stage MASEM, examining CBT's effect on youth depression via negative cognition. I hypothesized that the pooled correlations and path coefficients estimated from one-stage MASEM would not differ depending on whether Cohen's *d* or Hedges' *g* was used when the average total sample size of studies exceeds 100. This was grounded on two findings by Lin and Aloe (2020) and Lin (2018). First, Lin and Aloe's (2020) simulation research suggested that Hedges' *g* can yield additional bias compared to Cohen's *d* in meta-analytic contexts, but only in meta-analyses with studies with relatively small to moderate sample sizes. Second, Lin (2018) found out that the common utility of Hedges' g, bias correction for small sample sizes, diminishes when the average sample size of studies converges to 100 in meta-analyses. Similar to pooling SMDs to derive a meta-effect in meta-analytic studies, the MASEM methodology also pools the correlation coefficients after they are transformed from SMDs, particularly Cohen's *d* or Hedges' *g* in this case. Due to this similarity across methods, Lin and Aloe's (2020) and Lin's (2018) findings were anticipated to be evident in a MASEM analysis. Consequently, similar path coefficients and pooled correlations were expected in MASEM, regardless of whether the point-biserial correlation coefficients derived from Cohen's *d* or Hedges' *g*.

Indeed, the data analyses showed that the differences between the pooled correlations and path coefficients were decimally minimal – ranging from third to fourth decimal place – and had no practical impact on the interpretation of the model. Furthermore, both models showed the same patterns regarding significance, directionality, and magnitude. Hence, the results supported my hypothesis that Cohen's *d* and Hedges' *g* yielded equivalent outcomes when transformed into

point biserial correlation via equal-groups conversion (Equation (7)). Then, under my initial hypothesis, I conclude that the selection between Cohen's *d* and Hedges' *g* do not change the outcomes of one-stage MASEM, specifically for a meta-analysis of RCTs, with an average total sample size of approximately 101, examining the efficacy of CBT treatment on depression in youngsters through negative cognition.

Interpretation of the Findings

Although Lin and Aloe's (2020) and Lin's (2018) studies found that the pooled effect sizes, particularly Cohen's *d* and Hedges' *g*, do not yield their typical bias-related discrepancies when pooled in meta-analyses, my study conducts a different methodological approach. Unlike their studies, which solely focused on SMDs, this study additionally converts the SMDs into point-biserial correlation coefficients using the equal-groups conversion formula (Equation (7)). Moreover, the transformed SMDs (as well as the already-reported correlation coefficients) are pooled and later fed into MASEM analysis, which employs regression equations to illuminate (mediation) models within meta-analyses (Jak, 2015). Regardless of these differences, the data analysis revealed that the pooled correlations and path coefficients remained practically the same, irrespective of whether the correlation coefficients were derived from Cohen's *d* or Hedges' *g*. This accordance might be due to the fact that the average sample size in this study is near 100 (approximately 101), which is greater than the sample sizes in Lin and Aloe's (2020) and Lin's (2018) simulations, where they found a significant discrepancy in biases between Cohen's *d* and Hedges' *g*. Therefore, while the findings support the hypothesis and contribute to closing the gap in the literature that has received limited attention to date, they do not prove that the choice between Cohen's *d* and Hedges' *g* is practically negligible.

Strengths and Limitations

My study's main strength is that it bridges the gap between the theoretical differences between the most commonly used SMDs and their practical implications within the meta-analytical RCT framework via MASEM. Furthermore, by comparing these SMDs, this study gives insights into regularly overlooked methodological choices for researchers who conduct mediation meta-analyses in RCT contexts using SMDs, transformed point-biserial correlations, and MASEM. These methodological decisions are the choice of effect size estimator (particularly Cohens' *d* vs Hedges' *g*), the assumptions of the transformation formula that is used

to convert SMDs into point-biserial correlations, the aggregation method to handle dependent effect sizes and varying sample sizes within the same studies. Moreover, while there is research on how SMDs differ within meta-analytic frameworks, limited research has compared SMDs in the MASEM framework, which requires these SMDs to be transformed and pooled. Thus, this research sheds light on this gap by analyzing how SMDs' impact on results differs in MASEM contexts.

On the other hand, this study has several limitations. First, this study focused on only one dataset, where its average sample size might be too large to see any practical differences between Cohens' *d* and Hedges' *g.* The current analysis used a relatively large sample size of approximately 101. Even though Lin and Aloe (2020) and Lin (2018) showed that the bias correction of Hedges' *g* might not be observed in meta-analyses, the discrepancy in bias they found was pronounced when sample sizes were below approximately 30 to 70. Although the initial hypothesis is confirmed, it is important to consider that the discrepancy between Cohen's *d* and Hedges' *g* in this study might be less pronounced than in meta-analytic datasets with a smaller total sample size per study, particularly in the MASEM framework.

Second, because the current study focused on a single dataset with a specific mediation model, differences between the two SMDs may be more prominent in other mediation models or dataset characteristics, even with large sample sizes. For instance, some groups in the dataset were not perfectly balanced, akin to real-world RCT conditions. Several studies had imbalances between the control and treatment arms. While only a minority of the studies (12.50%) had perfectly balanced group sizes (i.e., equal numbers in treatment and control arms), the majority (52.71%) were nearly balanced, with base rates for treatment arms ranging from approximately 0.45 to 0.55. Thus, the results of my study may not be generalizable to other datasets with different (im)balances between arms or other mediation models.

Third, this study did not account for variations in sample sizes across studies. Although the average total sample size was approximately 101, the total sample sizes per study in the dataset were highly diverse. While most sample sizes (53%) were below 50, a non-negligible proportion of the sample sizes (21%) was above 200. In the former case, both Lin and Aloe (2020) and Lin (2018) showed differences in bias between Cohen's *d* and Hedges' *g*, indicating that the two MASEM models would have performed differently. In the latter case, however, both studies demonstrated that Cohen's *d* and Hedges' *g* yield the same bias as the sample size

exceeds 100. Such opposition might have canceled out any possible differences between Cohen's *d* and Hedges' *g* that would have been observed otherwise. Thus, the variations across sample sizes may not be generalizable to other datasets where sample sizes can be more consistent or diversely distributed.

Implications

Overall, the findings suggest that the choice between Cohen's *d* and Hedges' *g*, when transformed into point-biserial correlations using equal-samples conversion formula (Equation (7)), does not practically affect the outcomes of one-stage MASEM, particularly path coefficients and pooled correlations in the context of a meta-analysis of RCTs investigating CBT's effect on youth depression outcome via negative cognition. This result might be relevant for researchers employing meta-analytic methods to assess the effectiveness of psychological interventions (where SMDs are commonly used), and dependent effect sizes within the same studies exist.

Moreover, these findings have practical implications for researchers interested in using MASEM in psychological research, particularly in meta-analyses of RCTs. Since both SMDs yield equivalent results when converted into point-biserial correlations via the equal-samples conversion formula (Equation (7)) that assumes a balanced group (a principle inherent to RCT rationale), researchers can confidently use either effect size estimator in MASEM, particularly when the sample size is considerably large. Within this context, a sample size can be considered large when it exceeds 100, according to Zhang et al. (2013), in parallel with my study's sample size — approximately 101. Given that the differences between the two SMDs are negligible in practice, effect sizes from multiple studies (regardless of whether they use Cohen's *d* or Hedges' *g*) can be compared and synthesized. Then, for instance, mental health practitioners or policymakers can utilize these findings to understand the mechanisms of CBT in youths better to address the youth depression rates, which are on the rise (Wilson & Dumornay, 2022).

Future Directions

However, the generalizability of the current findings is limited, and future studies should account for these limitations. The first important direction could be to use meta-analytic datasets (or simulations) with smaller per-study sample sizes within the MASEM framework. Since the mean total sample size in the current study (approximately 101) exceeded the range where Lin (2018) and Lin & Aloe (2020) found the most pronounced differences between Cohen's *d* and

Hedges' *g*, and their focus was not on the MASEM frameworks, it would be valuable to replicate the current analysis using datasets with smaller sample sizes. Such work could help determine whether the practical equivalence between MASEM results regarding Cohen's *d* and Hedges' *g* is more pronounced as the sample sizes decrease.

Second, future (simulation) studies should systematically try different datasets with varying group balances and variations of sample size distributions. A simulation study or an empirical analysis could systematically compare the degree of imbalance and/or group size and how they affect the discrepancy in the results when the point-biserial correlations are derived from Cohen's *d* versus Hedges' *g.* Ultimately, studies investigating the difference between Cohen's *d* and Hedges' *g* with multiple aspects, such as different sample sizes or group imbalances, can help better generalize (or challenge) this study's findings, drawing a clearer picture of how Cohen's and Hedges' g differ from each other within the context of meta-analyses or RCTs and MASEM techniques.

This study examined whether using Cohen's *d* or Hedges' *g*, when transformed into point-biserial correlations, yields different outcomes in one-stage MASEM. Based on a meta-analytic dataset of RCTs examining the effect of CBT on youth depression through negative cognition, the results showed that both SMDs produced almost the same pooled correlations and path coefficients. These findings support the hypothesis that, under conditions of large sample sizes and balanced group designs, the choice between Cohen's *d* and Hedges' *g* is inconsequential for MASEM. Thus, researchers can use different effects from various studies in meta-analysis, whether these effects are derived from Cohen's *d* or Hedges' *g*. While the current study addresses a gap in the literature, further research should be conducted to test the generalizability of these conclusions under different circumstances, such as varying datasets, sample sizes, or balance conditions between treatment and control arms.

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Appendix A

The following script is used to preprocess the data files to compute Cohen's *d*, transform it to point-biserial correlation, and preprocess the data to produce a WebMASEM-compatible file in comma-separated values (CSV) format.

```
# 0- Import necessary libraries
library(dplyr)

# 1- Import datasets
ESdata <- read.csv(file.choose()) # Select ESdata.csv from local
file directory or paste the file path into read.csv()
unimeta<- read.csv(file.choose()) # Select unimeta.csv from
local file directory or paste the file path into read.csv()

# 2- Extract the CBT study IDs from the unimeta.csv (i.e., IPT
column = 0)
cbtStudyIDs <- unimeta %>%
  filter(IPT == 0) %>%  # Select only CBT trials (Study 1)
  pull(studyid) %>%  # Extract study IDs
  unique()  # Keep unique study IDs

# 3- Extract the relevant effect size data of CBT studies from
ESdata.csv based on mediation path and time of measurement
#     -Mediation paths (as shown in Table 1):
#       - No time-point CBT (0) → Post-treatment Negative
Cognition (3)
#       - Post-treatment Negative Cognition (3) → Post-treatment
Depression (3)
#       - No time-point CBT (0) → Post-treatments Depression (3)
ESDataCBTStudies <- ESdata %>%
  filter(
```

```
    studyid %in% cbtStudyIDs &
      (
        (var1type == "1" & var2type == "MA" & Var1time == 0 &
Var2time == 3) |    # Path a: CBT Treatment (1) → Negative
Cognition (MA)
          (var1type == "MA" & var2type == "O" & Var1time == 3 &
Var2time == 3) |  # Path b: Negative Cognition (MA) → Depression
Outcome (O)
          (var1type == "1" & var2type == "O" & Var1time == 0 &
Var2time == 3)     # Path c': CBT Treatment (1) → Depression
Outcome
      )
  ) %>%
  select(studyid, var1type, Var1time, var2type, Var2time,
mean_t, SD_t, n_t, mean_c, SD_c, n_c, r, NumOppDir,n_r, TCid,
var1mul, var2mul) # Only include parameters required to compute
Cohen's d

# 4- Compute Cohens d & reverse it's sign if the relationship
has 1 variable that is in the opposite direction, else same
direction.
ESDataCohensD <- mutate (ESDataCBTStudies,

  #Compute Cohen's d
  cohens_d = ((mean_t - mean_c) / (sqrt(((SD_t^2) * (n_t - 1) +
(SD_c^2) * (n_c - 1)) / (n_t + n_c - 2)))),

  #Reverse score cohens d that are comprised of 1 variable that
is in the opposite direction, otherwise same direction
  cohens_d_opp = ifelse((NumOppDir==1), -cohens_d, cohens_d),
```

```
  #Reverse score r that is comprised of 1 variable that is in
the opposite direction, otherwise same direction
  r = ifelse((NumOppDir==1), -r, r)
)


#5- Based on the code of meta-analysis (see ESComputation.R),
reverse r if one of the conditions below applies
#   -If var1type=1 AND var2type=Group A: O, MA, ML2, ML29
#   -If var1type=O, MA, ML2, or ML29 AND var2type=MB, MC, MD, ME
#   -If var1type=MB, MC, MD, ME AND var2type=O, MA, ML2, or ML29
ESDataCohensDSignsReversed <- mutate (ESDataCohensD,
                                      var1grp =
ifelse((var1type=="O"|var1type=="MA"|var1type=="ML2"|var1type=="
ML29"), "A", "B"),
                                      var2grp =
ifelse((var2type=="O"|var2type=="MA"|var2type=="ML2"|var2type=="
ML29"), "A", "B"),
                                      cohens_d =
ifelse((var1grp==var2grp), cohens_d_opp, -cohens_d_opp),
                                      r =
ifelse((var1grp==var2grp), r, -r),
)


# 6- Aggregate sample size and Cohen's d in case same
relationship within a study exists with different effect sizes
and/or sample sizes
ESDataAggregated <- ESDataCohensDSignsReversed %>%
  group_by(studyid, var1type, var2type) %>%
  summarise(
```

```
    cohens_d_avg = mean(cohens_d, na.rm = TRUE), # Average
Cohen's d (for relationships that may appear more than once per
study)
    r_avg = mean(r, na.rm = TRUE),               # Average r
(fallback when Cohen's d is missing, only happens in continuous
bivariate pairs)
    n_total_avg = mean(n_t + n_c, na.rm = TRUE)  # Average total
sample size (for same relationships with different sample sizes)
  ) %>%
  ungroup() %>%
  mutate(
    # Compute rpb from Cohen's d via Aaron et al. (1998). If
Cohen's d not reported, fallback to reported r (which is the
case for continuous relationship of M <--> Y)
    rpb = ifelse(!is.na(cohens_d_avg), (cohens_d_avg /
sqrt(cohens_d_avg^2 + 4 - (8 / n_total_avg))), r_avg)
    )

# 6- Compute webMASEM-compatible dataset with one row per study
webMASEMCohensD <- ESDataAggregated %>%
  mutate(
    # Recode paths
    M_X = ifelse(var1type == "1"  & var2type == "MA", rpb, NA),
# CBT → Negative Cognition
    Y_X = ifelse(var1type == "1"  & var2type == "O",  rpb, NA),
# CBT → Depression
    Y_M = ifelse(var1type == "MA" & var2type == "O",  rpb, NA)
# Negative Cognition → Depression
  ) %>%

  # Here, taking the mean does not take the mean, it just
collapses all the aggregated data into one row.
```

```
  group_by(studyid) %>%
  summarise(
    M_X = if (all(is.na(M_X))) NA else mean(M_X, na.rm = TRUE),
    Y_X = if (all(is.na(Y_X))) NA else mean(Y_X, na.rm = TRUE),
    Y_M = if (all(is.na(Y_M))) NA else mean(Y_M, na.rm = TRUE),
    N   = mean(n_total_avg, na.rm = TRUE)
  ) %>%
  ungroup()

# Write file into local directory within output folder.
if (!dir.exists("output")) dir.create("output")
write.csv(webMASEMCohensD, "output/webMASEMCohensD.csv")
```

Appendix B

The following script is used to preprocess the data files to compute Hedges' *g*, transform it to point-biserial correlation, and preprocess the data to produce a WebMASEM-compatible file in comma-separated values (CSV) format.

```
# Import necessary libraries
library(dplyr)

# 1- Import datasets
ESdata <- read.csv(file.choose()) # Select ESdata.csv from local
file directory or paste the file path into read.csv()
unimeta<- read.csv(file.choose()) # Select unimeta.csv from
local file directory or paste the file path into read.csv()

# 2- Extract the CBT study IDs from the unimeta.csv (i.e., IPT
column = 0)
cbtStudyIDs <- unimeta %>%
  filter(IPT == 0) %>%  # Select only CBT trials (Study 1)
  pull(studyid) %>%  # Extract study IDs
  unique()  # Keep unique study IDs

# 3- Extract the relevant effect size data of CBT studies from
ESdata.csv based on mediation path and time of measurement
#     -Mediation paths (as shown in Table 1):
#       - No time-point CBT (0) → Post-treatment Negative
Cognition (3)
#       - Post-treatment Negative Cognition (3) → Post-treatment
Depression (3)
#       - No time-point CBT (0) → Post-treatments Depression (3)
ESDataCBTStudies <- ESdata %>%
  filter(
    studyid %in% cbtStudyIDs &
```

```
      (
        (var1type == "1" & var2type == "MA" & Var1time == 0 &
Var2time == 3) |    # Path a: CBT Treatment (1) → Negative
Cognition (MA)
          (var1type == "MA" & var2type == "O" & Var1time == 3 &
Var2time == 3) |  # Path b: Negative Cognition (MA) → Depression
Outcome (O)
          (var1type == "1" & var2type == "O" & Var1time == 0 &
Var2time == 3)     # Path c': CBT Treatment (1) → Depression
Outcome
      )
  ) %>%
  select(studyid, var1type, Var1time, var2type, Var2time,
mean_t, SD_t, n_t, mean_c, SD_c, n_c, r, NumOppDir,n_r) # Only
include parameters required to compute Hedges' g

# 4- Compute Hedges g & reverse it's sign if the relationship
has 1 variable that is in the opposite direction, else same
direction.
ESDataHedgesG <- mutate (ESDataCBTStudies,

  #Compute Hedges'g
  hedges_g = (1 - (3 / (4 * (n_t + n_c) - 9))) * ((mean_t -
mean_c) / (sqrt(((SD_t^2) * (n_t - 1) + (SD_c^2) * (n_c - 1)) /
(n_t + n_c - 2)))),

  #Reverse score hedges_g that are comprised of 1 variable that
is in the opposite direction, otherwise same direction
  hedges_g_opp = ifelse((NumOppDir==1), -hedges_g, hedges_g),

  #Reverse score r that is comprised of 1 variable that is in
the opposite direction, otherwise same direction
```

```
  r = ifelse((NumOppDir==1), -r, r)
)


#5- Based on the code of meta-analysis (see ESComputation.R),
reverse r if one of the conditions below applies
#   -If var1type=1 AND var2type=Group A: O, MA, ML2, ML29
#   -If var1type=O, MA, ML2, or ML29 AND var2type=MB, MC, MD, ME
#   -If var1type=MB, MC, MD, ME AND var2type=O, MA, ML2, or ML29
ESDataHedgesGSignsReversed <- mutate (ESDataHedgesG,
                                var1grp =
ifelse((var1type=="O"|var1type=="MA"|var1type=="ML2"|var1type=="
ML29"), "A", "B"),
                                var2grp =
ifelse((var2type=="O"|var2type=="MA"|var2type=="ML2"|var2type=="
ML29"), "A", "B"),
                                hedges_g =
ifelse((var1grp==var2grp), hedges_g_opp, -hedges_g_opp),
                                r = ifelse((var1grp==var2grp), r,
-r),
)

# 6- Aggregate sample size and Hedges'g in case same
relationship within a study exists with different effect sizes
and/or sample sizes
ESDataAggregated <- ESDataHedgesGSignsReversed %>%
  group_by(studyid, var1type, var2type) %>%
  summarise(
    hedges_g_avg = mean(hedges_g, na.rm = TRUE), # Average
Hedges' g (for relationships that may appear more than once per
study)
```

```
    r_avg = mean(r, na.rm = TRUE),                # Average r
(fallback when Hedges' g is missing, only happens in continuous
bivariate pairs)
    n_total_avg = mean(n_t + n_c, na.rm = TRUE)  # Average total
sample size (used to fill in missing sample sizes later)
  ) %>%
  ungroup() %>%
  # Step: Derive rpb from hedges_g_avg when available; otherwise
fallback to r_avg
  mutate(
    # Compute rpb from Hedges' g via Aaron et al. (1998). If
Hedges' g not reported, fallback to reported r (which is the
case for continuous relationship of M <--> Y)
    rpb = ifelse(!is.na(hedges_g_avg), (hedges_g_avg /
sqrt(hedges_g_avg^2 + 4 - (8 / n_total_avg))), r_avg)
)

# 6- Compute webMASEM-compatible dataset with one row per study
webMASEMHedgesG <- ESDataAggregated %>%
  mutate(
    # Recode effect paths
    M_X = ifelse(var1type == "1"  & var2type == "MA", rpb, NA),
# CBT → Neg. Cognition
    Y_X = ifelse(var1type == "1"  & var2type == "O",  rpb, NA),
# CBT → Depression
    Y_M = ifelse(var1type == "MA" & var2type == "O",  rpb, NA)
# Neg. Cognition → Depression
  ) %>%

  # Here, taking the mean does not take the mean, it just
collapses all the aggregated data into one row per study.
  group_by(studyid) %>%
```

```
  summarise(
    M_X = if (all(is.na(M_X))) NA else mean(M_X, na.rm = TRUE),
    Y_X = if (all(is.na(Y_X))) NA else mean(Y_X, na.rm = TRUE),
    Y_M = if (all(is.na(Y_M))) NA else mean(Y_M, na.rm = TRUE),
    N   = mean(n_total_avg, na.rm = TRUE)
  ) %>%
  ungroup()

# Write file into local directory within output folder.
if (!dir.exists("output")) dir.create("output")
write.csv(webMASEMHedgesG, "output/webMASEMHedgesG.csv")
```

Appendix C

Below is the Lavaan syntax that was used to identify the model to the WebMASEM software and used in the meta-analytic structural equation modeling.

```
# Intercepts
x ~ 1;
m ~ 1;
y ~ 1;
# Regression paths
m ~ beta1*x;
y ~ beta2*m + b_yx*x;
# Variances
x ~~ 1*x;
m ~~ 1*m;
y ~~ 1*y;
```